

UMAquizz1.pdf



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Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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Pregunta 1

Completa

Puntuació 1,00
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pregunta

In a UMA (Uniform Memory Access time) multiprocessor, there are two or more identical processors connected to a single shared main memory through an interconnection network; this network allows any processor to access any memory location.

Which of the following statements are true?

In a UMA system, processors access to shared memory using instructions that are different from those used to access local memory (i.e. cache).

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

🚩 Marca la
pregunta

In a UMA system with private (local) caches, the time to access main memory on a miss is independent of which processor is doing the access and which memory address is being accessed.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals

WUOLAH

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00

 [Marca la pregunta](#)

In a UMA system with private (local) caches, the coherence protocol guarantees that for each line in main memory there can only exist a single copy of the line in one of the private cache memories of the system, independently of the number of processors available.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

Pregunta 4

Completa

Puntuació 1,00
sobre 1,00

 [Marca la pregunta](#)

In a UMA system with private (local) caches, a write-update coherence policy ensures that all copies in private caches of a memory line are updated.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals

Pregunta 5

Completa

Puntuació 1,00
sobre 1,00

 [Marca la pregunta](#)

In a UMA system with private (local) caches and write-update coherence, the access to a memory address previously accessed by the same processor always results in a cache hit, unless the cache replacement algorithm decided to replace the line that contains that address.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals



WUOLAH

Pregunta 6

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

In a UMA system with write-invalidate MSI, the BusRdX invalidation command is generated on the bus every time a processor performs PrWr on a valid line in its cache memory.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

Pregunta 7

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

In a UMA system with write-invalidate MSI, when a processor performs a PrRd followed by a PrWr to an address residing in an uncached (or invalid) memory line, the associated snoopy needs to place two coherence commands on the bus, even when no other cores have accessed the memory location in between.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals

Pregunta 8

Completa

Puntuació 1,00
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In a UMA system with write-invalidate MSI, the BusRdX does not need to read the memory line that contains the address being accessed since the processor asks for the line with the purposes of modifying (writing to) it.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

**WUOLAH**

Pregunta 9

Completa

Puntuació 1,00
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In a UMA system with the simplest write-invalidate MSI protocol explained in class, main memory is responsible for providing a memory line when one of the private caches asks for it unless the line is in M state in another private cache.

Trieu-ne una:

- ☒ Vertader
- ☐ Fals

Pregunta 10

Completa

Puntuació 1,00
sobre 1,00 [Marca la pregunta](#)

The implementation of the simplest write-invalidate MSI protocol explained in class requires two bits to track the state of each word in each cache line. Therefore, if the cache has capacity for N lines, each one storing M words, MSI requires $N \cdot M \cdot 2$ bits to keep its contents coherent.

Trieu-ne una:

- ☐ Vertader
- ☒ Fals

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